Euclid, Book I, Proposition I.10

To Bisect a Given Finite Straight Line

CGJteamLab

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1 Introduction

Proposition I.10 constructs the midpoint of a given finite straight line.

In Euclid's development this construction depends on several earlier results. An equilateral triangle is first erected on the given segment using Proposition I.1. The vertex angle of this triangle is then bisected by Proposition I.9, and the Side–Angle–Side congruence criterion of Proposition I.4 is used to prove that the intersection point is the midpoint of the original segment.

The Hilbert reconstruction follows a fundamentally different approach. The existence of a midpoint has already been established as an independent theorem of the Hilbert interface. Consequently, Proposition I.10 becomes a direct application of this previously derived result.

This proposition therefore provides one of the clearest examples of the difference between Euclid's historical development and a modern axiomatic reconstruction.

1.1 Initial Configuration

Let AB be the given finite straight line.

The objective is to construct a point lying on the segment AB that divides it into two congruent parts.

Such a point is called the midpoint of the segment.

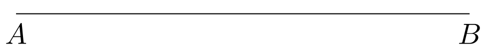

Figure 1: The given finite straight line AB.

1.2 Construction

Construct the equilateral triangle ABC on the segment AB by Proposition I.1.

Next, bisect the angle $\angle ACB$ by Proposition I.9.

Let the angle bisector meet the segment AB at the point D.

The claim is that D is the midpoint of the segment AB.

1.3 Application of Proposition I.4

Consider the triangles

$$\triangle ACD \quad \text{and} \quad \triangle BCD.$$

Since the triangle ABC is equilateral,

$$AC \cong BC.$$

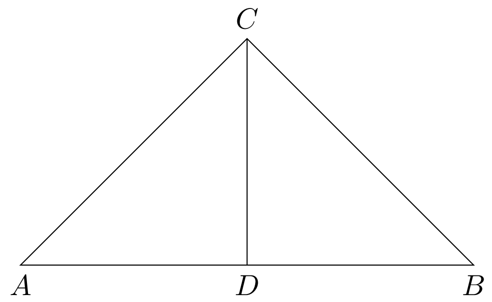

Figure 2: Euclid's construction of the midpoint of a segment.

The segment

$$CD$$

is common to both triangles, and by construction,

$$\angle ACD \cong \angle DCB.$$

Therefore the hypotheses of Proposition I.4 are satisfied.
It follows that

$$AD \cong DB.$$

Since the point D lies on the segment AB, it is the midpoint of the given finite straight line.

2 Statement

Theorem 1 (Euclid I.10)**.** *Let AB be a finite straight line with*

$$A \neq B.$$

Then there exists a midpoint of the segment AB. That is, there exists a point M such that

$$A, M, B$$

are collinear,

$$M$$

lies between A and B, and

$$AM \cong MB.$$

Equivalently, every nondegenerate segment admits a midpoint.

3 Proof

Construct the equilateral triangle ABC on the segment AB by Proposition I.1.
Bisect the angle $\angle ACB$ by Proposition I.9, and let the bisecting ray meet the segment AB at the point D.
Since

$$AC \cong BC,$$

$$CD \cong CD,$$

and

$$\angle ACD \cong \angle DCB,$$

the triangles

$$\triangle ACD \quad \text{and} \quad \triangle BCD$$

satisfy the hypotheses of Proposition I.4.
Hence

$$AD \cong DB.$$

Because the point D lies on the segment AB, it is the midpoint of the given finite straight line.

This completes the construction.

4 Implementation Architecture

The Hilbert reconstruction replaces the entire classical construction by a previously established theorem of the Hilbert interface.

Rather than constructing an auxiliary equilateral triangle, bisecting an angle, and applying the Side–Angle–Side criterion, the existence of a midpoint is obtained directly from the general midpoint existence theorem.

Consequently, Proposition I.10 becomes a simple wrapper around an existing interface theorem.

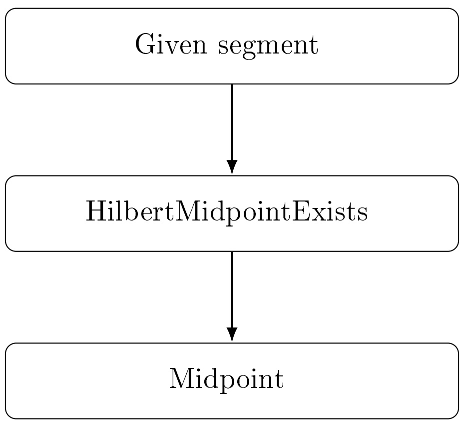

Figure 3: Architecture of the Hilbert reconstruction of Proposition I.10.

5 Hilbert Reconstruction

The reconstruction consists of a direct application of the general midpoint existence theorem.

Given a nondegenerate segment, the theorem `HilbertMidpointExists` immediately produces a point lying between the endpoints and equidistant from them.

No auxiliary geometric construction is required, since the existence of midpoints has already been established independently within the Hilbert development.

This represents one of the largest simplifications obtained by the layered architecture of the Hilbert interface.

6 Lean Formalization

The Lean implementation mirrors the Hilbert reconstruction exactly.

After assuming that the endpoints of the given segment are distinct, the proof consists of a single application of the theorem

`HilbertMidpointExists.`

This theorem immediately returns a point lying between the endpoints and equidistant from them, precisely satisfying the definition of a midpoint.

Consequently, the formal proof contains no auxiliary constructions and makes no explicit use of Propositions I.1, I.4, or I.9. All of the classical geometric reasoning has already been incorporated into the previously established Hilbert interface.

7 Discussion

Proposition I.10 provides one of the clearest illustrations of the difference between Euclid's historical development and a layered axiomatic reconstruction.

In the *Elements*, the midpoint construction depends on three earlier propositions. Euclid first constructs an equilateral triangle, then bisects one of its angles, and finally applies the Side–Angle–Side criterion to prove that the resulting point divides the original segment into two equal parts.

The Hilbert reconstruction is fundamentally different. The existence of midpoints has already been established as an independent theorem of the Hilbert interface. Consequently, Proposition I.10 becomes a direct application of this theorem, requiring no additional geometric construction.

This proposition demonstrates the principal advantage of the layered architecture developed in CGJteamLab. Once sufficiently rich interface theorems have been established, many classical propositions cease to be independent proofs and instead become concise applications of the existing theory.

Proposition I.10 is perhaps the simplest example of this philosophy, reducing an entire classical construction to a single theorem call.